
Vector Algebra

Vector Addition

$$Out[*]= \begin{pmatrix} -8 \\ -9 \\ 6 \\ 4 \end{pmatrix} + \begin{pmatrix} 5 \\ -1 \\ -2 \\ -9 \end{pmatrix} = \begin{pmatrix} -3 \\ -10 \\ 4 \\ -5 \end{pmatrix}$$

Scalar Multiplication

$$Out[*]= -3 \cdot \begin{pmatrix} 1 \\ 5 \\ 9 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ -15 \\ -27 \\ -6 \end{pmatrix}$$

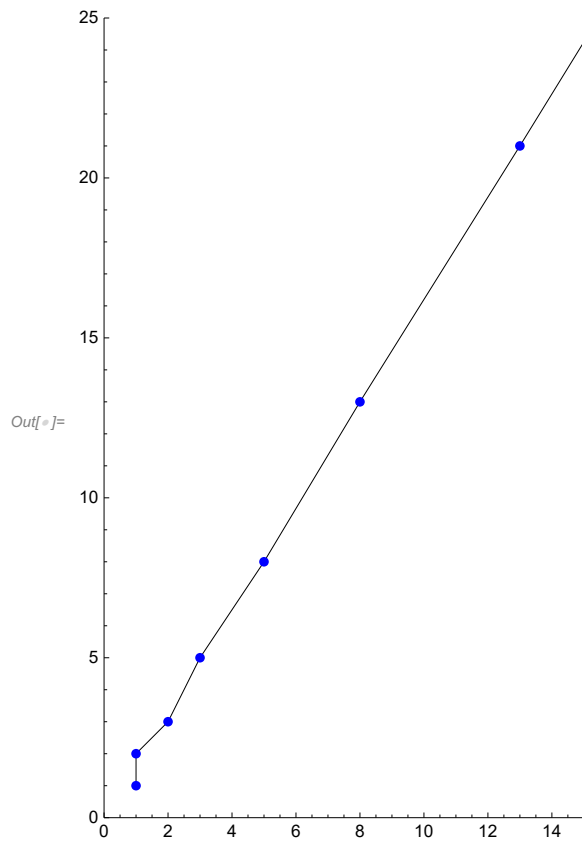
Vector Dynamics

Fibonacci

$$Out[*]= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x+y \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} 2 \\ 3 \end{pmatrix} \rightarrow \begin{pmatrix} 3 \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} 5 \\ 8 \end{pmatrix} \rightarrow \begin{pmatrix} 8 \\ 13 \end{pmatrix} \rightarrow \begin{pmatrix} 13 \\ 21 \end{pmatrix}$$

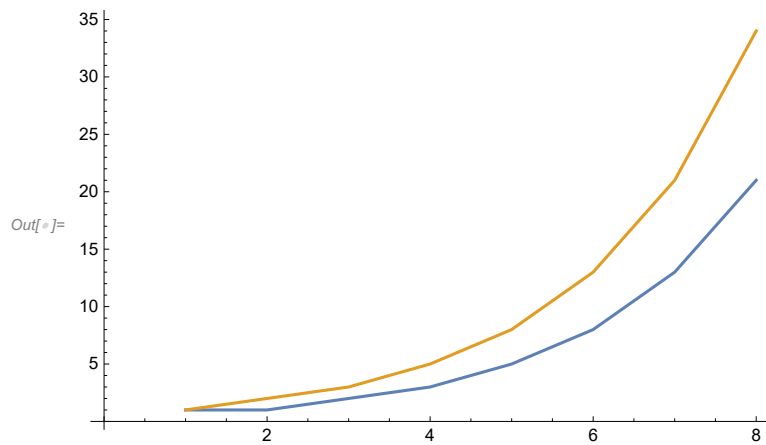
Fibonacci



It becomes a straight line.

The ratio of one segment to the next approaches a fixed number.

Fibonacci

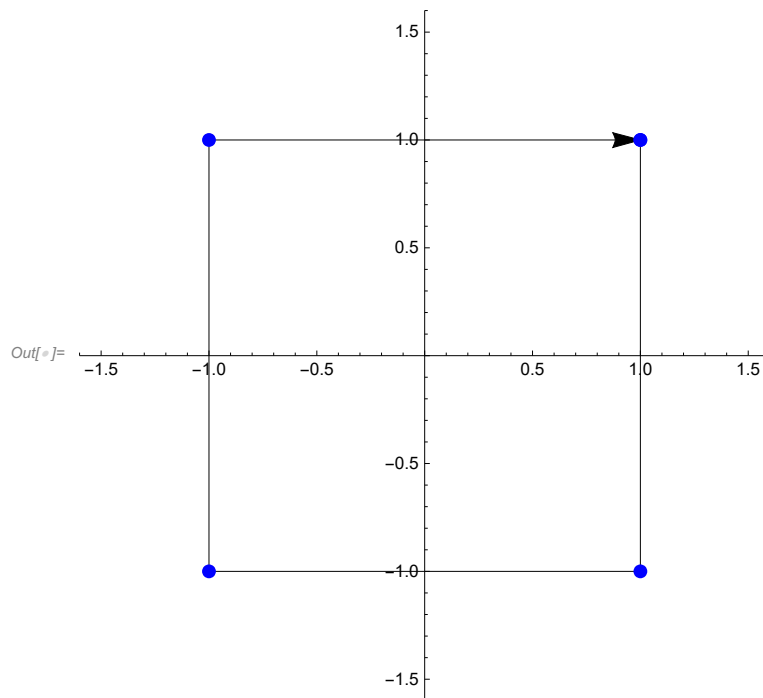


Rotation

$$\text{Out}[*]= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

$$\text{Out}[*]= \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

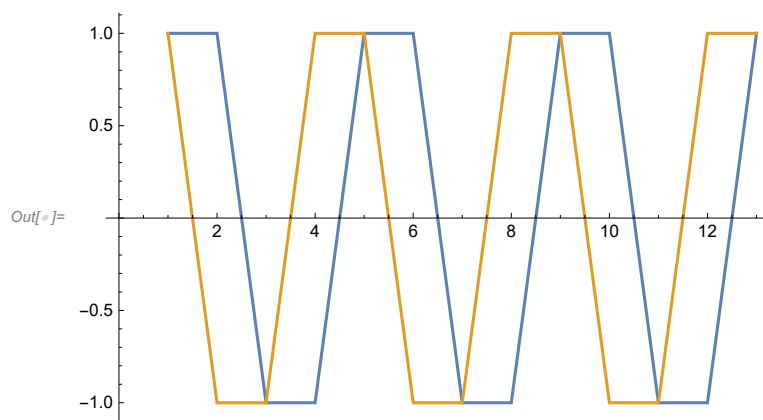
It repeats.



It rotates like the symmetries of the square.

The ratio between segments is one.

Rotation



Matrices

Matrix Product

$$\text{Out}[*]= \begin{pmatrix} -7 & 8 \\ -7 & -4 \end{pmatrix} \cdot \begin{pmatrix} -3 & -7 \\ 7 & -3 \end{pmatrix} = \begin{pmatrix} 77 & 25 \\ -7 & 61 \end{pmatrix}$$

Matrix Product

$$\text{Out}[38]= \begin{pmatrix} -3 & 6 \\ 9 & -5 \end{pmatrix} \cdot \begin{pmatrix} -3 & -3 \\ -8 & 2 \end{pmatrix}$$

$-3 \cdot -3 + 6 \cdot -8$	$-3 \cdot -3 + 6 \cdot 2$
$9 \cdot -3 + -5 \cdot -8$	$9 \cdot -3 + -5 \cdot 2$

$$\begin{pmatrix} -3 \cdot -3 + 6 \cdot -8 & -3 \cdot -3 + 6 \cdot 2 \\ 9 \cdot -3 + -5 \cdot -8 & 9 \cdot -3 + -5 \cdot 2 \end{pmatrix}$$

$$\begin{pmatrix} 9 + -18 & 24 + 12 \\ -27 + 15 & -72 + -10 \end{pmatrix}$$

$$\begin{pmatrix} -9 & 36 \\ -12 & -82 \end{pmatrix}$$

Matrix Product

$$\text{Out}[*]= \begin{pmatrix} -7 & 4 \\ -3 & -4 \end{pmatrix} \cdot \begin{pmatrix} -3 & -6 \\ 4 & -9 \end{pmatrix} = \begin{pmatrix} 37 & 6 \\ -7 & 54 \end{pmatrix}$$

Matrix Product

$$\text{Out}[*]= \begin{pmatrix} 7 & 0 \\ 5 & 8 \end{pmatrix} \cdot \begin{pmatrix} 8 & 2 \\ 4 & -1 \end{pmatrix} = \begin{pmatrix} 7 \cdot 8 + 0 \cdot 4 & 7 \cdot 2 + 0 \cdot -1 \\ 5 \cdot 8 + 8 \cdot 4 & 5 \cdot 2 + 8 \cdot -1 \end{pmatrix} = \begin{pmatrix} 56 & 14 \\ 72 & 2 \end{pmatrix}$$

Matrix Product

$$\text{Out}[*]= \begin{pmatrix} 1 & 2 & -9 \\ 2 & 7 & -1 \end{pmatrix} \cdot \begin{pmatrix} -8 & -7 & 7 & -8 \\ -1 & -7 & 8 & -8 \\ 5 & -5 & 7 & 9 \end{pmatrix} = \begin{pmatrix} -55 & 24 & -40 & -105 \\ -28 & -58 & 63 & -81 \end{pmatrix}$$

Linear Algebra Theory

Linear Transformations

A linear transformation is a map T between vector spaces.

$T(v + w) = T(v) + T(w)$ preserves vector addition

$T(a v) = a T(v)$ preserves scalar multiplication

Matrix Multiplication with a vector is a linear transformation

$$T(v) = A \cdot v$$

All linear transformations can be written as a matrix.

Any vector space has a basis

The standard basis vectors are those with one one in them.

$$\text{Out[28]= } b_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad b_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

A basis does not need to be the standard basis.

Any vector space has a basis

Any vector is a linear combination of the basis vectors.

$$\begin{pmatrix} 3 \\ 5 \\ 7 \\ 4 \end{pmatrix} = 3 b_1 + 5 b_2 + 7 b_3 + 4 b_4$$

A basis is linearly independent

$$\text{if } \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0 = c_1 b_1 + c_2 b_2 + c_3 b_3 + c_4 b_4 = \sum c_i b_i \text{ then } c_i = 0 \text{ for all } i$$

Dimension

There are many choices of basis vectors, but they all have the same number of basis vectors.

The number of basis vectors is called the dimension.

$$\text{Out[]= } 7 = \text{dimension of the span of } \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Null Space

The null space of a matrix is the set of vectors which it maps to the zero vector.

$$A \cdot v = 0$$

Null Space

For example, suppose

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x + 2y = 0$$

$$2x + 4y = 0 \quad \text{we get } x = -2y$$

$$\begin{pmatrix} -2y \\ y \end{pmatrix} \text{ where } y \text{ can be anything}$$

The null space is a vector space.

Image

The image is the set of vectors w such that there is a solution v to

$$w = A \cdot v$$

The image is a vector space.

Matrix Rank

The dimension of the image is called the rank of the matrix.

If the matrix is n by m , it has n rows and m columns.

The dimension of the null space plus the rank of the matrix is equal to the number of columns.

Square matrices

If the number of rows equals the number of columns, it is a square matrix.

$$\text{Out[8]=} \quad \begin{pmatrix} 9 \end{pmatrix} \quad \begin{pmatrix} 9 & -4 \\ 4 & -8 \end{pmatrix} \quad \begin{pmatrix} 6 & -1 & 4 \\ 4 & -2 & -3 \\ -1 & -9 & -4 \end{pmatrix} \quad \begin{pmatrix} 6 & -7 & -5 & 6 \\ 4 & 6 & -3 & -8 \\ 1 & -5 & -5 & 0 \\ 8 & -9 & 7 & 9 \end{pmatrix}$$

A square matrix can be thought of as a linear transformation from a vector space to itself.

Eigenvalue and Eigenvector

It makes sense to compare a vector in the domain and in the range for a square matrix.

$$\text{Out[39]=} \quad \begin{pmatrix} 7 & 0 \\ -2 & 3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7x \\ -2x + 3y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

Eigenvalue and Eigenvector

There is a better way, but let's try solving this system of equations.

$$7x = \lambda x$$

$$-2x + 3y = \lambda y$$

Eigenvalue and Eigenvector

$$\begin{aligned} 7x &= \lambda x \quad \text{that implies either } x \text{ is zero or } \lambda = 7 \\ -2x + 3y &= \lambda y \end{aligned}$$

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Eigenvalue and Eigenvector

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Eigenvalue and Eigenvector

There are two possibilities.

$$\begin{pmatrix} 0 \\ y \end{pmatrix} \text{ with eigenvalue } \lambda = 3$$

$$\begin{pmatrix} -2y \\ y \end{pmatrix} \text{ with eigenvalue } \lambda = 7$$

Identity matrix

There is one special square matrix, called the identity matrix.

$$\text{Out[]:= } \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

It has ones on the diagonal.

Identity matrix

It is the matrix version of the standard basis.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Identity matrix

It is called the identity matrix because it does not change the vector.

Identity matrix

There is one eigenvalue $\lambda = 1$,

but we say it has multiplicity two because the eigenspace is two dimensional.

In general, the eigenvalue of the identity has multiplicity the same as the dimension.

Diagonal Matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x \\ -y \\ 4z \end{pmatrix}$$

Diagonal Matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x \\ -y \\ 4z \end{pmatrix}$$

Eigenvalue $\lambda = 2$ with eigenvector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Diagonal Matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x \\ -y \\ 4z \end{pmatrix}$$

Eigenvalue $\lambda = -1$ with eigenvector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Diagonal Matrix

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x \\ -y \\ 4z \end{pmatrix}$$

Eigenvalue $\lambda = 4$ with eigenvector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Products of Diagonal Matrices

Try it and see that you just multiply each entry separately.

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 10 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -8 \end{pmatrix}$$

Diagonal matrices commute. $A \cdot B = B \cdot A$

Powers of Diagonal Matrices

Remember how to get the general formula for $f(n) = r \cdot f(n - 1)$

$$f(n) = c \cdot r^n$$

Powers of Diagonal Matrices

For vector dynamics, we might have a matrix A instead of r : $f(n) = A \cdot f(n - 1)$

$$f(n) = A^n \cdot f(0)$$

Powers of Diagonal Matrices

For a diagonal matrix, the power can be done on each entry separately.

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}^n = \begin{pmatrix} 2^n & 0 & 0 \\ 0 & (-1)^n & 0 \\ 0 & 0 & 4^n \end{pmatrix}$$

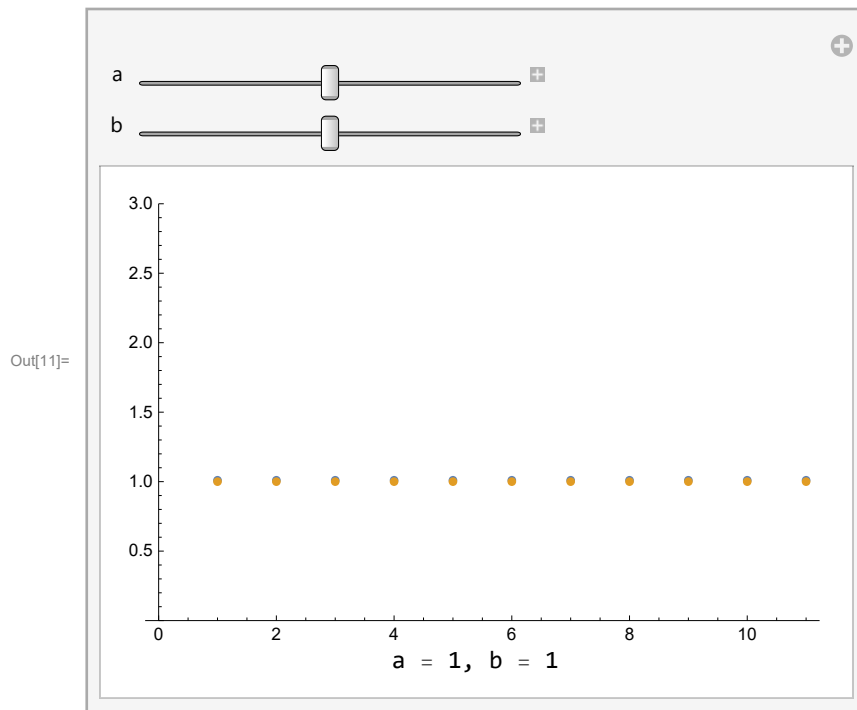
Powers of Diagonal Matrices

Let's visualize the dynamics for the diagonal matrix with coefficients a and b

$$f(n) = A \cdot f(n - 1) \text{ and } f(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} a^n & 0 \\ 0 & b^n \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a^n \\ b^n \end{pmatrix}$$

Powers of Diagonal Matrices : $\begin{pmatrix} a^n \\ b^n \end{pmatrix}$



Determinant

The determinant is for square matrices only. rows = columns

The determinant of a diagonal matrix is the product of the entries on the diagonal.

$$\det \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix} = 2 \cdot -1 \cdot 4 = -8$$

Determinant

The determinant of a 2 by 2 matrix is

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \cdot d - b \cdot c$$

Determinant

$$\det \begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix} = 3 \cdot 5 - 7 \cdot 2 = 15 - 14 = 1$$

The product of the eigenvalues (with multiplicity) is the determinant.

Matrix inverse

The matrix inverse is for square matrices only. rows = columns

$$\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix} \cdot \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 15 - 14 & -21 + 21 \\ 10 - 10 & -14 + 15 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

If $A \cdot B = I$ then B is the inverse of A

it also follows that $B \cdot A = I$ and A is the inverse of B

In general $A \cdot B$ is not the same as $B \cdot A$

Things you should know and be able to do

Multiply and add matrices and vectors (whenever it is appropriate).

Solve systems of linear equations.

Find the eigenvalues and eigenvectors of a 2x2 matrix

Know about the identity matrix.

Know about the zero vector.

Intermediate thing to know before eigenvectors

How to compute the null space of a matrix. (More general than eigenvector)

How to compute the determinant of a 2 by 2 matrix (More general than eigenvalue)

Use the quadratic formula (in the process of computing eigenvalues)